

Problems: Cox-Gorenstein Algebras

Cox-Gorenstein Algebras, Toric Macaulay-Matlis Duality, and Lefschetz Properties in Toric Varieties

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Abstract

We would like to investigate the structural properties of Cox-Gorenstein algebras, a special class of Artinian algebras graded by finitely generated abelian groups that arise naturally in the study of Cox rings of toric varieties. In a recent work in collaboration with R. Holanda and W. Montoya [1] we recall the Macaulay-Matlis duality in the toric context, the codimension one conjecture of Cattani-Cox-Dickenstein, and the toric Lefschetz properties (weak and strong), obtaining a generalized Hessian criterion for these properties. The results obtained have potential applications in algebra and geometry.

Keywords: Cox-Gorenstein algebras, Toric varieties, Cox ring, Macaulay-Matlis duality, Lefschetz properties, Hessian criterion, Codimension one conjecture.

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1 Introduction

The theory of graded Artinian algebras has deep roots in classical commutative algebra, dating back to Macaulay's seminal work on inverse systems and Gorenstein algebras [7]. In recent decades, there has been a resurgence of interest in this area, driven by deep connections with algebraic geometry, Hodge theory, algebraic combinatorics, and intersection theory.

Graded Artinian algebras appear naturally in various mathematical contexts:

- **Toric geometry:** The Cox rings of toric varieties provide natural examples of algebras graded by finitely generated abelian groups that are not simply $\mathbb{Z}$ -graded [5].
- **Lefschetz theory:** The weak and strong Lefschetz properties for Artinian algebras have been intensively studied over the past decade, with applications in Hodge theory, intersection theory, and algebraic combinatorics [8].
- **Residue theory:** The codimension one conjecture of Cattani-Cox-Dickenstein, proposed in 1997, remains open in its full generality and represents a fundamental challenge in toric residue theory [2].

Despite recent advances, several fundamental questions remain unanswered:

1. **Complete characterization of Cox-Gorenstein algebras:** Although we know that these algebras satisfy Poincaré duality, a complete classification based on combinatorial properties of the grading group has not been established.
2. **Codimension one conjecture:** The conjecture of Cattani-Cox-Dickenstein [2] remains open in its full generality, despite significant progress in particular cases [4, 3].
3. **Toric Lefschetz properties:** The Lefschetz theory for G -graded algebras with G a finitely generated abelian group is still in its infancy, with many open problems.

2 Cox-Gorenstein algebras, an overview

2.1 Toric Varieties and Cox Rings

Definition 2.1 (Toric Variety). A toric variety is an irreducible variety X containing an algebraic torus $T \simeq (\mathbb{C}^*)^d$ as a Zariski open subset such that the action $T \times T \rightarrow T$ of T on itself extends to an algebraic action of T on X .

Definition 2.2 (Cox Ring of a Toric Variety). Let X be a complete toric variety. The **Cox ring** of X is the graded ring

$$\text{Cox}(X) := \bigoplus_{[D] \in \text{Cl}(X)} H^0(\mathcal{O}_X(D)) \quad (1)$$

where $\text{Cl}(X)$ denotes the class group of X .

If $X = \mathbb{P}_\Sigma$ is a complete toric variety, then its Cox ring $S = \text{Cox}(X)$ is a polynomial ring with one variable for each ray of the fan Σ . The natural grading is given by the class group $\text{Cl}(X)$, which is a finitely generated abelian group.

Remark 2.3. The converse also holds: every finitely generated abelian group can be presented as the class group of some toric variety. Moreover, given a finitely generated abelian group G and a G -grading over a polynomial ring S , we can recover the toric variety when the rank of G is less than or equal to the number of variables of S .

2.2 Artinian G -Graded Algebras

Let $(G, +, \preceq)$ be a partially ordered abelian group, $\mathbb{K}$ a field of characteristic zero, and R a $\mathbb{K}$ -algebra.

Definition 2.4 (G -Graded Algebra). A G -grading on R is a decomposition of R as a direct sum of $\mathbb{K}$ -vector spaces

$$R = \bigoplus_{g \in G} R_g \quad (2)$$

such that the product in R satisfies $R_g R_h \subset R_{g+h}$. We always consider $R_0 = \mathbb{K}$ and say that R is finitely graded if $R_g = 0$ for all but finitely many $g \in G$. Given a homogeneous element $f \in R_g \setminus \{0\}$, we denote $\deg(f) = g \in G$. We say that the algebra is an Artinian G -graded algebra if it is an Artin ring, neglecting the grading.

To properly encode the structural information contained in the Hilbert function, we introduce the Hasse-Hilbert diagram.

Definition 2.5 (Hilbert Function and Hasse-Hilbert Diagram). Let A be an Artinian G -graded $\mathbb{K}$ -algebra. The Hilbert function of A is

$$\text{HF}_A : G \rightarrow \mathbb{Z}_+, \quad \text{HF}_A(g) = h_g := \dim_{\mathbb{K}} A_g. \quad (3)$$

The Hasse-Hilbert diagram of A is defined as a vertex-weighted directed graph structure over the covering graph of G , where the weight of a vertex g is its Hilbert function h_g .

Definition 2.6 (Extremal Elements). By definition of the covering graph of a partial order, its vertex set is the poset G , and two vertices $g, h \in G$ are connected if they are immediate neighbors, that is, they are comparable and there is no other comparable element between them. A maximal element in a poset X is an element $x \in X$ for which there does not exist $y \in X$ such that $x \preceq y$ with $x \neq y$. We say that a maximal element $x \in X$ is the greatest element in X if $y \preceq x$ for all $y \in X$.

2.3 Cox-Gorenstein Algebras

Definition 2.7 (Cox-Gorenstein Algebra). Let $A = \bigoplus_{g \in G} A_g$ be an Artinian G -graded $\mathbb{K}$ -algebra of finite type. We say that A is **Cox-Gorenstein** if there exists $\rho \in G$ such that

$$\text{Soc}(A) := (0 : \mathfrak{m}) = A_\rho \simeq \mathbb{K} \quad (4)$$

where $\mathfrak{m}$ denotes the maximal ideal of A .

Remark 2.8. Notice that the socle of an arbitrary G -graded Artinian algebra is always a maximal element in the poset. On the other hand, when the algebra has the greatest element, then its degree is called the socle degree of the algebra. This is always the case when A is Cox-Gorenstein.

Definition 2.9 (Poincaré Duality Algebra). Let $A = \bigoplus_{g \in G} A_g$ be an Artinian G -graded $\mathbb{K}$ -algebra of finite type. We say that A has **Poincaré duality** if there exists a maximal element $\rho \in G$ such that $A_\rho \simeq \mathbb{K}$ and the multiplication maps

$$A_g \times A_{\rho-g} \rightarrow A_\rho \quad (5)$$

are perfect pairings whenever A_g and $A_{\rho-g}$ are non-zero. Moreover, if one is zero, then the other is also zero.

Theorem 2.10 (Poincaré Duality Characterization). *Let A be a G -graded Artinian $\mathbb{K}$ -algebra of finite type. Then A is Cox-Gorenstein if and only if A has the Poincaré duality.*

Let $S = \mathbb{K}[x_1, \dots, x_s]$ be the polynomial ring with a positive G -grading. Let $Q = \mathbb{K}[X_1, \dots, X_s]$ be the ring of differential operators associated to S , that is, $X_i = \frac{\partial}{\partial x_i}$, and S has a natural structure of Q -module given by differentiation $X_i(x_j) = \delta_{ij}$. The G -grading on S induces a G -grading on Q by defining $\deg(X_i) = \deg(x_i) \in G$.

Remark 2.11. Notice that the action has descending degree, that is, for $a \preceq b$, we have $Q_a \times S_b \rightarrow S_{b-a}$.

Definition 2.12 (Annihilator and Inverse System). Given a G -graded Q -submodule of S , $M \subset S$, we define the annihilator

$$\text{Ann}(M) = \{\alpha \in Q \mid \alpha(f) = 0 \text{ for all } f \in M\}. \quad (6)$$

Just as in the classical case, $\text{Ann}(M) \subset Q$ is an ideal and it is easy to see that it is G -homogeneous. Conversely, given a G -homogeneous ideal $I \subset Q$, we define the inverse system

$$I^{-1} = \{f \in S \mid \alpha(f) = 0 \text{ for all } \alpha \in I\}. \quad (7)$$

Again, $I^{-1} \subset S$ is a G -graded Q -submodule of S .

Theorem 2.13 (Toric Macaulay-Matlis Duality). *There is a bijective correspondence*

$$\begin{array}{ccc} \{G\text{-Homogeneous ideals } I \subset Q\} & \longleftrightarrow & \{G\text{-Graded } Q\text{-submodules } M \subset S\} \\ I & \longrightarrow & I^{-1} \\ \text{Ann}(M) & \longleftarrow & M \end{array} \quad (8)$$

Under this correspondence, $M = I^{-1}$ is finitely generated as an S -module if and only if $A = Q/I$ is Artinian. Moreover, A is Artinian Gorenstein if and only if $M = Q \cdot f$ is a cyclic module.

Theorem 2.14 (Macaulay's Theorem for Cox-Gorenstein Algebras). *Let $I \subset Q = \mathbb{K}[X_1, \dots, X_n]$ be a G -homogeneous ideal such that $A = Q/I = \bigoplus_{g \in G} A_g$ is an Artinian G -graded $\mathbb{K}$ -algebra. Then A is Cox-Gorenstein of socle degree ρ if and only if there exists $f \in S_\rho$ such that $I = \text{Ann}(f)$.*

2.4 The Codimension One Conjecture

Definition 2.15 (Critical Degree). Let X be a complete simplicial toric variety and let $f_0, \dots, f_n \in B(\Sigma)$ be homogeneous polynomials which do not vanish simultaneously on X , with $\deg(f_i) = \alpha_i$. Let β be the sum of the degrees of the variables; it is well known that β is the anticanonical class on X . The critical degree of this configuration is

$$\rho = \sum_{i=0}^d \alpha_i - \beta. \quad (9)$$

Conjecture 2.16 (Cattani-Cox-Dickenstein [2]). *Let $A = S/(f_0, \dots, f_n)$ be an algebra in this setting. Then*

$$\dim_{\mathbb{C}} A_{\rho} = 1. \quad (10)$$

Remark 2.17. In the same paper, the authors prove that the codimension one conjecture implies that A_{ρ} is a maximal element, that is, A_{ρ} is part of the socle of A .

2.5 Toric Lefschetz Properties

Let $A = Q/I = \bigoplus_{g \in G} A_g$ be a G -graded Artinian $\mathbb{K}$ -algebra, and let $\mathcal{L} = \langle X_1, \dots, X_n \rangle \subseteq A$ be the $\mathbb{K}$ -linear subspace generated by the classes of the variables in Q . We denote $\mathcal{L}_g = \mathcal{L} \cap A_g$.

Definition 2.18 (Toric Strong Lefschetz Property). We say that A has the **Toric Strong Lefschetz Property** (TSLP) if for every homogeneous A_g and A_h such that $h = g + k\ell$, there exists a linear element $L \in \mathcal{L}_{\ell}$ such that the $\mathbb{K}$ -linear multiplication map

$$\bullet L^k : A_g \rightarrow A_h \quad (11)$$

has maximal rank.

Definition 2.19 (Toric Weak Lefschetz Property). We say that A has the **Toric Weak Lefschetz Property** (TWLP) if for every linearly consecutive pieces A_g and A_h , there exists a linear element $L \in \mathcal{L}_{h-g}$ such that the $\mathbb{K}$ -linear multiplication map

$$\bullet L : A_g \rightarrow A_h \quad (12)$$

has maximal rank.

2.6 Toric Hessian Criterion

Definition 2.20 (Toric Mixed Hessian). Let $\mathcal{B} = \{\beta_1, \dots, \beta_s\}$ and $\mathcal{C} = \{\gamma_1, \dots, \gamma_t\}$ be $\mathbb{K}$ -bases of A_{g_i} and A_{g_j} , respectively. The toric mixed Hessian of f with mixed order $(\mathcal{B}, \mathcal{C})$ is

$$\text{Hess}_f^{(\mathcal{B}, \mathcal{C})} := [\beta_i \circ \gamma_j(f)]. \quad (13)$$

Remark 2.21. As usual, the definition of the Hessian depends on the basis, but its rank does not.

Theorem 2.22 (Toric Hessian Criterion). *Let $A = Q/I$ with $I = \text{Ann}(f)$ be an Artinian G -graded Cox-Gorenstein $\mathbb{K}$ -algebra of socle degree $\rho \in G$. Let A_g and A_h be two linearly comparable graded pieces of A such that $g < h$ and $h = g + k\ell$ with $L = a_1X_1 + \cdots + a_mX_m \in \mathcal{L}_\ell$. Then the matrix of the $\mathbb{K}$ -linear map $\bullet L^k : A_g \rightarrow A_h$ with respect to the bases $\mathcal{B}$ and $\mathcal{C}$ is:*

$$[\bullet L^k]_{\mathcal{B}}^{\mathcal{C}} = k! \cdot \text{Hess}_f^{(\mathcal{C}^*, \mathcal{B})}(a) \quad (14)$$

where $a = (a_1, \dots, a_m, 0, \dots, 0)$. In particular, these Hessians control the TSLP.

3 Problems

3.1 Cox-Gorenstein monomial algebras

We know that Artinian Gorenstein monomial algebras are always Artinian complete intersections:

$$A = k[x_1, \dots, x_n]/(x_1^{a_1}, \dots, x_n^{a_n}).$$

In the standard graded setup, we know that these algebras have the Strong Lefschetz Property (SLP) as soon as $\text{char}(k) = 0$.

Problem: Assuming $\text{char}(k) = 0$ and G a finitely generated abelian group, consider S with a G -grading and

$$A = S/(x_1^{a_1}, \dots, x_n^{a_n}).$$

Is it true that A has TSLP/TWLP in this case?

3.2 Codimension two Cox-Gorenstein

In the standard graded case, all Artinian Gorenstein algebras of codimension two have the SLP.

Problem: What can be said in the G -graded case?

3.3 A higher Jacobian criterion and Jordan types

Investigate whether a higher Jacobian-type criterion can be established in the G -graded setting.

3.4 Artinianization in the critical degree

Let $A = S/(f_0, \dots, f_n)$ be an algebra in the setting of Catani-Cox and assume that the codimension one conjecture is true for A . Let B be the algebra given by the natural artinianization truncating on the critical degree. Is B a Cox-Gorenstein algebra?

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